

Week 7: Fast Computation of Prices, Greeks, Implied Vol, and Surfaces

The latency budget, in C++ via pybind11 – Milestone 2

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Learning Objectives

- ▶ Implement characteristic-function pricers in C++ (COS, Carr-Madan)
- ▶ Implement robust implied-vol inversion and surface revaluation
- ▶ Expose the C++ kernels to Python with pybind11

Lecture 1: CF Pricing & Implied Vol in C++

- ▶ Characteristic functions under Black-Scholes and Heston
- ▶ The Fang-Oosterlee COS method (the workhorse)
- ▶ Robust implied-vol inversion: good guess + safeguarded Newton
- ▶ Vectorizing a whole surface

Lecture 2: Binding C++ to Python with pybind11

- ▶ How a pybind11 module is structured and built with CMake
- ▶ Passing NumPy/PyTorch tensors with no copy (buffer protocol)
- ▶ Releasing the GIL around the C++ hot loop
- ▶ PyTorch autograd is the REFERENCE for Greeks; C++ is the speed

Milestone 2 Deliverable

- ▶ C++ pricing/IMPLIED-VOL/Greeks library, imported via pybind11
- ▶ Full BTC surface (300-500 contracts) revalued under 5 ms
- ▶ Greeks within $1e-4$ of PyTorch autograd; pricing within $1e-4$ of reference
- ▶ Latency reported as a distribution, plus a BINDING.md explainer